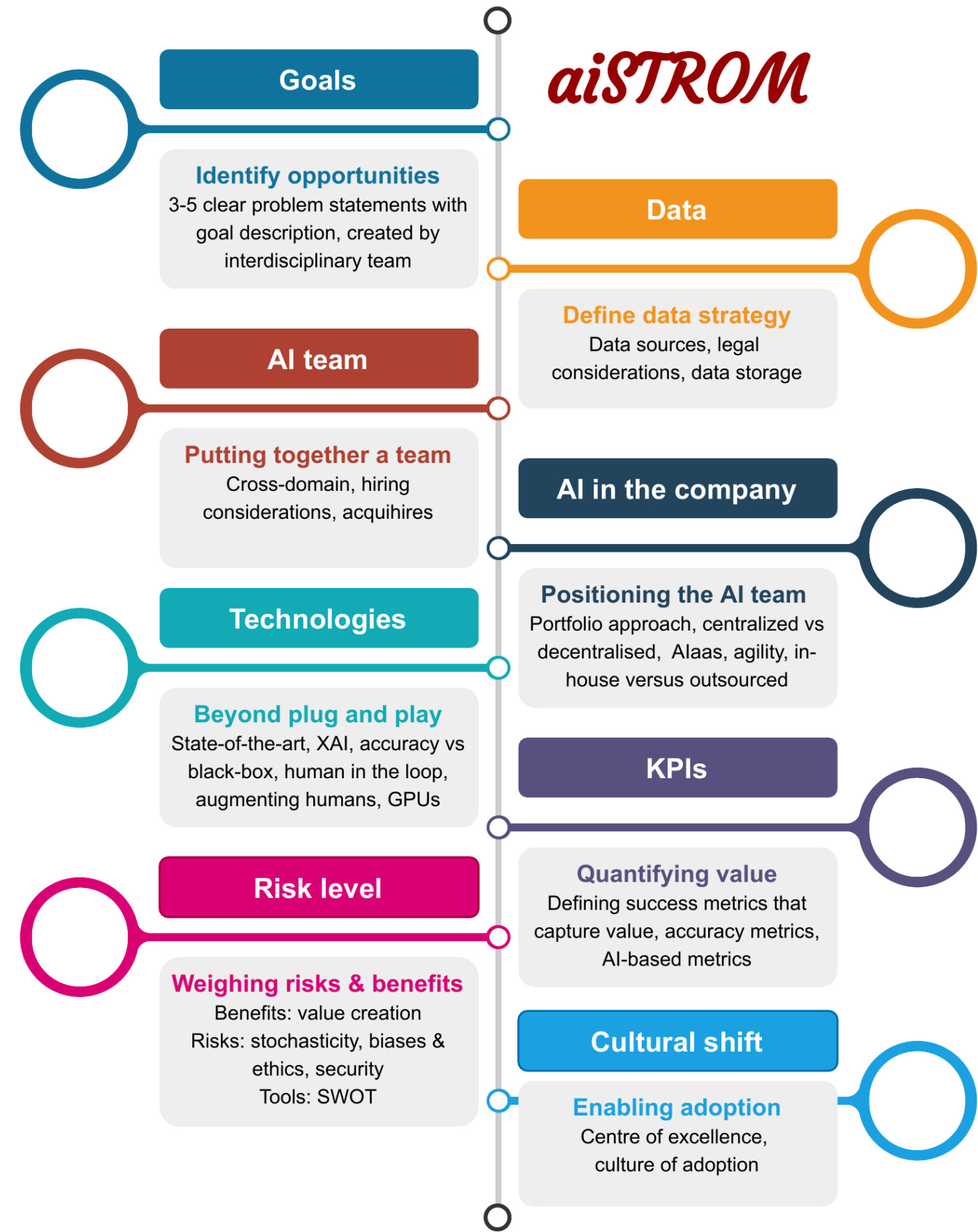


Decide which AI projects to pursue, and what each one requires, before choosing a solution.



How to use these sheets

- 1 Before you start: build an interdisciplinary team**
Combine people with domain and organisation knowledge (e.g. C-level and employees) with people who have AI knowledge (e.g. AI consultants or developers).
- 2 Sheet 1, Goals: identify opportunities**
List possible goals and problem statements that AI could tackle. Select the top n (typically 3-5), maximising potential impact for the lowest effort. Give them letters A, B, C.
- 3 Sheets 2-7: analyse each shortlisted project**
Work through Data, AI team, AI in the company, Technologies, KPIs and Risk level. Each sheet has one column per project, so the differences between projects become visible. End each column with a conclusion.
- 4 Sheet 8, Cultural shift**
Plan the education and adoption the projects need: AI literacy, a centre of excellence, continuous learning.
- 5 Sheet 9, Decide**
Plot the projects by risk and benefit, run a SWOT per project and choose a balanced portfolio that fits the organisation's risk strategy. Uncertain projects usually start with a prototype.

ORGANISATION / STRATEGIC QUESTION
FICTIONAL EXAMPLE. A 1,200-person company with offices in Singapore, Malaysia and Germany. Question: should we use AI in our employee expense-claims process, and for what?
TEAM (DOMAIN AND AI KNOWLEDGE)
Finance director (sponsor), two Finance reviewers, internal audit, HR, IT lead, privacy officer, external AI consultant.
STARTING SITUATION
About 2,500 expense claims a month. Employees photograph each receipt and then retype merchant, date, amount and category (about 8 minutes per claim). Five Finance staff check every claim by hand; reimbursement takes 11 days on average. All figures are illustrative.
DATE / NEXT REVIEW
September 2026. Next review: January 2027, after the first prototype.

3-5 clear problem statements with goal description, created by an interdisciplinary team

WHERE TO LOOK FOR IDEAS

Internal efficiency gaps

Where could AI make the organisation perform more smoothly and efficiently?

Employees retype every receipt (about 330 staff hours a month). Finance checks every claim manually; reimbursement takes 11 days.

New AI technologies

Which new technologies exist, and how could they be of use to us?

Receipt reading (OCR and extraction) is now a mature, established task. Anomaly detection can flag unusual claims.

Competitors

What are competitors doing? What should we emulate, or what are we competing with?

Peers, and our own expense-software vendor, advertise auto-filled claims. Staff expect it from consumer apps.

New business models

Are new business models on the rise that could replace our current services?

No new business model threatens us here. This is an internal process opportunity, not a new product.

#	PROBLEM STATEMENT / GOAL DESCRIPTION	HOW AI CREATES VALUE	IMPACT	EFFORT	TOP n
1	Auto-fill claims from receipt photos: merchant, date, amount, currency, category.	Automate processing	High	Medium	A
2	Flag duplicate, altered or unusual claims for Finance to review.	Predictions	High	High	B
3	Check claims against the expense policy before they are submitted.	Automate processing	Medium	Medium	C
4	Dashboard of spending trends by department, supplier and trip.	Find trends in data	Medium	Low	-
5	Chatbot that answers employees' questions about the expense policy.	Automate processing	Low	Medium	-
6	Predict which claims will be delayed in approval.	Predictions	Low	Medium	-
7		-	-	-	-
8		-	-	-	-

AI can create value by 1) automating processing, 2) identifying trends from historic data, or 3) making intelligent predictions that empower people. Choose the top n (typically 3-5) with the highest impact for the lowest effort and give them letters A, B, C; these become the columns of sheets 2-9.

WHY THESE PROJECTS? WHAT HAPPENS TO THE OTHER IDEAS?

Top n = 3. A has the highest impact for moderate effort and affects every employee. B protects money and can reuse the data that A produces, but is harder. C is cheaper and partly solvable without AI, so it is a useful comparison. Idea 4 does not need AI: it goes to the business-intelligence team. Ideas 5 and 6 are parked: low impact for their effort.

Data sources, legal considerations, data storage

	PROJECT A	PROJECT B	PROJECT C
	Receipt auto-fill	Fraud & duplicate flags	Policy pre-check
Data sources Does the organisation already have the necessary data? Is big data needed, or will more data not help (curse of dimensionality)? What should we start collecting now? III-B.1	About 90,000 receipt images (3 years) with the amounts Finance approved: usable as ground truth. Not big data, and volume is sufficient. Start storing employee corrections now.	Only about 400 confirmed duplicates or fraud cases in 3 years: very few positive examples. Start recording every flag and its outcome now; the data will grow slowly.	The policy is a PDF that changed twice in 3 years. Past rejections and their reasons sit in free-text comments, not in structured fields.
Collection vs acquisition Collect it ourselves (time, ethics approval, quality checks), buy it, or annotate it (experts or crowdsourcing)? Garbage in, garbage out. III-B.1	No need to buy data. Only if we train our own model: Finance experts label about 2,000 receipts (about 60 hours). No crowdsourcing: receipts are confidential.	Fraud data matching our claims cannot be bought. Label past audit findings with Finance and internal audit; accept that collection takes 1-2 years.	Encode the policy as rules with Finance. Label about 500 past rejections by reason to test the rules.
Data characteristics Volume, velocity, variety, veracity (bias, noise) and value: what does each mean for this project? III-B	Variety is the challenge: 4 currencies, 3 languages, phone photos of crumpled receipts. Veracity is good: approved amounts were checked by Finance.	Veracity is weak: past checks were inconsistent, so "not flagged" does not mean "clean". Expect noisy labels and missed cases in the history.	Rejection reasons are inconsistent free text and need cleaning before use.
Legal: privacy & security Which privacy laws apply (e.g. GDPR, PDPA, CCPA; every country we operate in)? Who is the privacy manager? Balance easy internal access against breaches and exfiltration. III-B.2	Receipts show names, hotels, locations and card digits. GDPR (German staff) and PDPA (Singapore, Malaysia) apply. The privacy officer signs off. Bank details stay in the payroll system.	Screening employees for possible fraud is sensitive under GDPR: it needs a legal basis, consultation of the German works council and transparency to staff.	Little extra risk: it uses claim fields that Finance already processes.
Storing data Internal infrastructure or external data centre? Stored close to its users? Data lake, warehouse or semi-structured? Who administers it? III-B.3	Images stay in the current expense system (EU and Singapore data centres). If an external service reads them, the contract must cover location, retention and no training on our data.	Flags and outcomes are stored in a restricted Finance and audit table; access limited to three named people.	Rules live in the expense system's configuration; no new storage.
Conclusion The data strategy for this project in one or two sentences.	Data is available. The main work is privacy approval and capturing employee corrections from now on.	Data is the bottleneck. Start collecting labelled outcomes now; build a model later.	Mostly rules. A small labelled set is enough for testing.

2026 note: when using hosted generative AI services, prompts and uploaded files are data too. Check where they are stored and whether the provider may train on them.

		PROJECT A	Receipt auto-fill	PROJECT B	Fraud & duplicate flags	PROJECT C	Policy pre-check
Skills needed Mathematics & statistics, data mining, big data engineering, pattern extraction, visualisation & UI, software engineering, communication, continuous learning. Which does this project need?	III-C		Software engineering and integration; evaluating extraction accuracy; designing a clear confirmation screen. No deep mathematics needed if we use an existing service.		Statistics and data mining for rare events, feature engineering with Finance, and communicating why a claim was flagged.		Software engineering to encode the rules. Little machine learning.
Domain knowledge Who brings the domain knowledge into the multidisciplinary team?	III-C		Two Finance reviewers define the fields, categories and what counts as an error.		Internal audit and Finance know the fraud patterns. HR joins because flags concern employees.		The Finance policy owner, with HR for exceptions.
Obtaining talent Up-skill existing employees (technical and vertical AI knowledge), hire (graduates, PhDs, academic collaboration) or acquihire?	III-C		IT has the integration skills. No hiring: the extraction capability is bought (see sheet 4).		We have no data scientist. Options: up-skill a Finance analyst with the external consultant, or hire one data scientist. An acquihire is not justified at our scale.		Existing IT developers.
Retaining talent How will we keep AI experts: continuous education, benefits, working environment? Where are we dependent on one person?	III-C		Low dependence on scarce AI talent. Document the integration so it survives staff changes.		One data scientist would be a key-person risk: pair them with the Finance analyst and fund training time.		Finance documents the rules so they survive staff changes.
Conclusion The team for this project, and the main gap.			Feasible with current staff plus a vendor.		Needs new skills: decide between up-skilling and hiring before starting.		Feasible with current IT.

Portfolio approach, centralised vs decentralised, Alaas, agility, in-house versus outsourced

	PROJECT A	PROJECT B	PROJECT C
	Receipt auto-fill	Fraud & duplicate flags	Policy pre-check
Positioning the team Centralised (hub, centre of excellence, chief AI officer), decentralised (spokes) or hub-and-spoke? Does the team have access to data across departments? Short reporting line? Who has decision power?	We have no central AI team. Finance owns the project (a spoke), IT integrates, the privacy officer signs off. Short reporting line to the Finance director.	Finance ownership, with internal audit and HR in the decision group because the outcomes affect employees.	Finance owns it; IT configures it. If more AI projects follow, consider a small central hub (hub-and-spoke).
Portfolio approach Short term (existing, low-risk technology), medium term (emerging technology) or long term (risky, competitive advantage)? Funded in internal rounds?	Short term: existing, low-risk technology. A quick win that builds trust for B.	Medium term: collect data now, prototype in 2027. Funded only if A succeeds and the data analysis shows patterns.	Short term; runs alongside A.
AI as a service (Alaas) Does a service already do this (do not reinvent the wheel)? Is it core technology for us? How exposed are we to price changes?	Receipt reading is not our core business: do not reinvent the wheel. Our expense software sells it as an add-on and two other services exist. Ask for a price guarantee.	Vendor fraud modules exist but are generic. Check whether they can learn from our own cases.	Our expense software already has a configurable rules engine.
In-house, outsource or acquire What is our AI readiness (pioneer, investigator, experimenter, passive)? Build in-house, outsource development, or grow by acquisition?	Our AI readiness: experimenter. Buy the service and build only the integration.	Start with the vendor module plus the consultant. Build in-house only if the data proves useful.	Configure in-house.
Agility Which prototype tests the idea cheaply? Which stakeholders and users to involve early? Will we need MLOps to bring models into production?	8-week prototype with 50 volunteers in all three countries before any contract. No MLOps needed while the vendor hosts the model.	First an exploratory analysis of past audits: does the data contain detectable patterns? Then a prototype. MLOps needed if built in-house.	Refine the rules with Finance every month.
Conclusion Where the project sits and how it will be developed.	Finance-led; buy and integrate; prototype first.	Finance and audit; data first; decide build or buy after exploration.	Configure existing software; no AI.

2026 note: Alaas now also means hosted foundation models such as LLMs. The same questions apply: dependence, price changes and what data leaves the organisation.

	PROJECT A	PROJECT B	PROJECT C
	Receipt auto-fill	Fraud & duplicate flags	Policy pre-check
State of the art & hybrid Which techniques fit the task? Is it an established task or something nobody has done? Would a hybrid with older AI (rules, fuzzy logic) be more efficient?	Receipt OCR and extraction is an established task. Combine it with simple rules, e.g. date and currency checks.	Hybrid: rules catch exact duplicates; anomaly detection looks for the rest.	Rules, not machine learning. AI only if free-text descriptions need classifying.
Accuracy vs black box (XAI) Must the outcome be explainable, legally or to the people affected? Would we accept lower accuracy for an interpretable model?	Little explanation needed: the employee sees the extracted fields next to the receipt and corrects them.	Must be explainable: Finance and the employee need the reason, e.g. "same amount and merchant as claim 1234". Prefer an interpretable model over a slightly more accurate black box.	Rules are fully explainable.
Human in the loop Where do ground-truth labels come from? Can people add new labels while the system is in production?	Every employee correction becomes a new label for improving the model.	Finance's decision on each flag becomes a label, so the model improves while in use.	Finance reviews the exceptions the rules raise.
Replacing or augmenting Should the system replace the person doing the task, or augment them?	Augment: removes typing; the employee still confirms every field.	Augment: AI flags, Finance decides. Never reject a claim automatically.	Augment: warns the employee before submission.
Cloud vs in-house hosting What computing (GPU) power is needed? Rent cloud servers or buy equipment (capex and opex)? Small edge devices?	Vendor cloud; no GPUs to buy. Check that data stays in the EU or Singapore.	Small models on tabular data: existing servers or cloud; no GPUs needed.	Existing system.
Conclusion The technical approach for this project.	Buy an extraction service; a person confirms each claim.	Rules plus an interpretable model; Finance decides on every flag.	Rules engine; no AI.

2026 note: generative models can produce fluent but wrong output. If a system can act on its own (an 'agent'), decide what it may do without human approval.

Defining success metrics that capture value, accuracy metrics, AI-based metrics

		PROJECT A	PROJECT B	PROJECT C
		Receipt auto-fill	Fraud & duplicate flags	Policy pre-check
Value-based KPIs Which KPIs link the project to strategic goals? Think of customers, costs, processes and investors, not only profit.	III-F	Process: minutes per claim (8 now, target 3). Employees as internal customers: satisfaction with claims. Cost: Finance hours per month.	Cost: money recovered or prevented. Process: share of flags that Finance judges justified.	Process: share of claims returned for policy reasons (9% now).
Hidden value What value is harder to measure: internal efficiency, better understanding of customers, better decisions?	III-F	Clean, structured spending data that makes idea 4 (spend analytics) easier. Goodwill from staff.	Deterrence: fewer attempts once staff know checks exist.	Fewer emails back and forth between Finance and staff.
Model performance metrics Which accuracy measures (accuracy, confusion matrix, F1) and which level is needed? When would we pivot?	III-F	Accuracy per field, per country and per language. Prototype target: at least 95% correct amounts. Pivot if below 85%.	Precision and recall from the confusion matrix. Finance can review at most 30 flags a month, so precision matters most.	Share of past rejections the rules would have caught.
AI-based metrics Could AI itself produce better or more insightful KPIs (e.g. sentiment analysis, business intelligence)?	III-F	Extracted categories feed automatic spend reports.	Fraud patterns by type become audit KPIs.	Classify free-text rejection reasons to see which rules confuse staff.
Reporting Who reviews the KPIs and how often, to justify continued investment?	III-F	Monthly to the Finance director during the prototype, quarterly afterwards.	Quarterly to the audit committee.	Monthly in the Finance team meeting.
Conclusion What success looks like, in numbers.		Claim time down 60% with at least 95% correct amounts.	At least half of all flags justified, and a measurable amount recovered.	Policy returns halved.

Benefits: value creation. Risks: stochasticity, biases & ethics, security

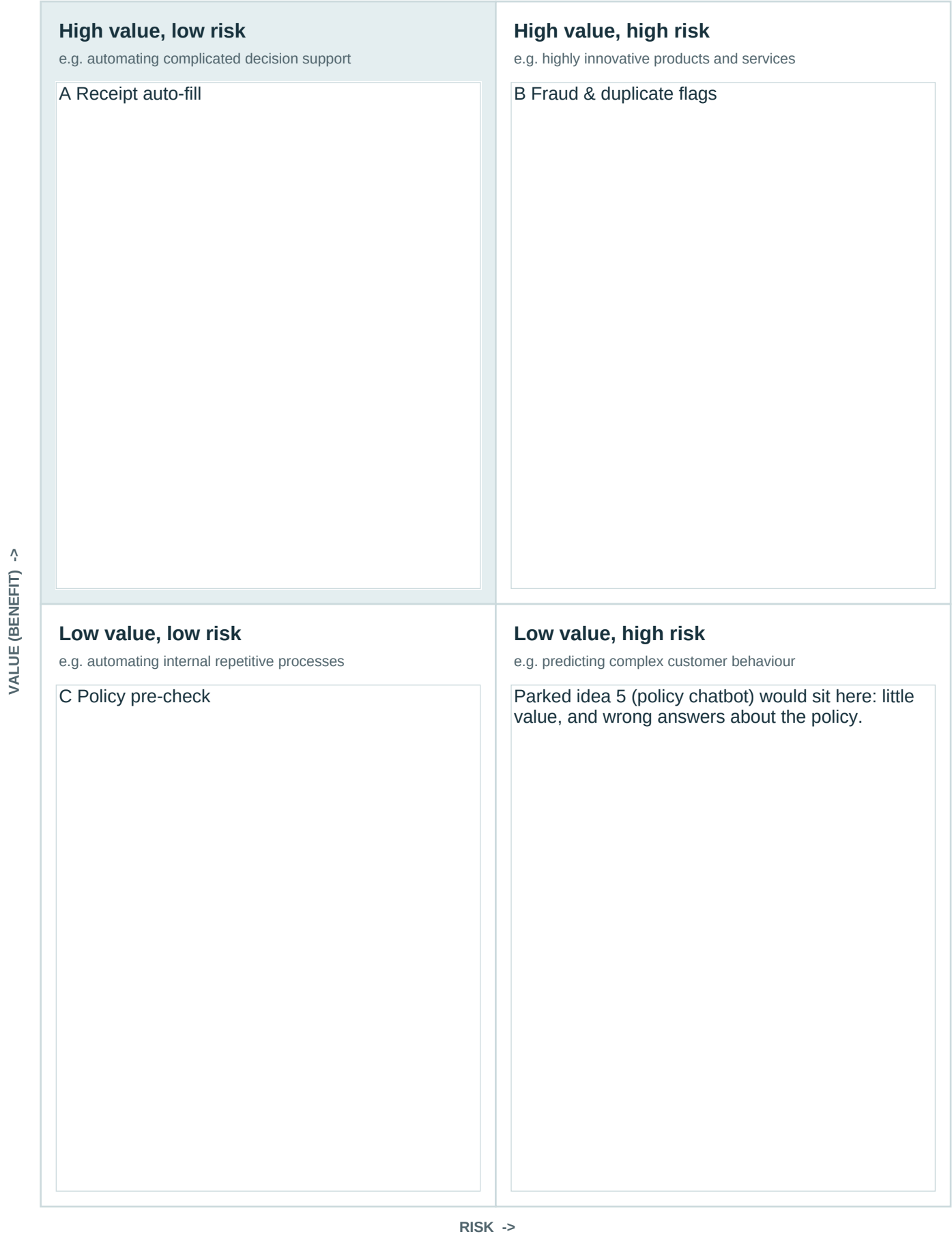
		PROJECT A	Receipt auto-fill	PROJECT B	Fraud & duplicate flags	PROJECT C	Policy pre-check
Stochastic nature There is no guarantee the model will work. How uncertain is feasibility? Do competitors have working models? Is a prototype needed first? What is the pivot?	III-G.1		Low: an established task, and vendors have working models. The open question is how well they read our receipts; the prototype answers it.		High: no guarantee our data contains detectable patterns. Exploratory analysis first, with a willingness to pivot.		Low: rules behave predictably.
Biases & ethics Could historical data teach the model unfair biases? Who could be treated unfairly? Which ethical dilemmas: let the algorithm decide or set rules?	III-G.1		Lower accuracy for some languages or receipt types would burden staff in one country. Test results per group.		High: past audits may have focused on some departments, and a model could learn this. HR reviews; appeal route; no automatic rejection.		The policy itself may treat some staff unfairly, e.g. part-time travellers. Review with HR.
Security Could deceptive inputs fool the model (adversarial attacks, spoofing)? What else needs protecting?	III-G.1		Adversarial input: edited or AI-generated fake receipts can fool extraction. Keep image and duplicate checks (links to B).		Fraudsters adapt to known checks (evasion attacks). Do not publish thresholds; review the model regularly.		Little new exposure.
Other strategic risks New or hard technology? In-house know-how? Price guarantee for Alaas? Team access to data? A new task or an established one?	III-G.1		Dependence on the vendor and its prices. Negotiate a price guarantee and the right to export our data.		A new task for us, and a key-person risk on one data scientist.		None significant.
Benefits What value does the project create, for customers or internally?	III-G.2		About 200 staff hours saved per month, faster reimbursement and better data.		Money protected, stronger controls and deterrence.		Fewer returned claims and a clearer policy.
Conclusion Overall risk level and benefit level (low / medium / high), for the plot on sheet 9.			Risk: low. Value: high.		Risk: high. Value: medium to high.		Risk: low. Value: low to medium.

2026 note: for generative AI, add hallucination (stochastic nature) and prompt injection (security).

	PROJECT A	PROJECT B	PROJECT C
	Receipt auto-fill	Fraud & duplicate flags	Policy pre-check
Educating managers & team What do the managers and developers directly involved need to know about AI?	Finance director and reviewers: half a day on what extraction can and cannot do, and how to read the accuracy results.	Audit and Finance: what precision, recall and false flags mean in practice.	Finance: how to maintain the rules.
AI literacy for everyone What do users of the system and other employees need to understand?	All staff: a 2-minute guide. Always check the filled-in fields; you remain responsible for your claim.	Tell staff openly that claims are screened and how flags are reviewed. This builds trust and meets GDPR transparency.	Short hints in the app.
From fear to empowerment Do employees fear being replaced? How will we show that AI can empower them, and how they work with the system?	Finance reviewers fear job loss. Commitment: freed time goes to supplier audits and analysis, not headcount cuts.	Staff may feel watched. Explain the purpose, offer an appeal route and involve the works council.	Little concern.
Centre of excellence Who creates awareness and knowledge of AI? Can employees join brainstorming workshops about new AI ideas?	Too small for a centre of excellence now. A Finance-IT-HR working group shares lessons and runs an AI idea workshop next year.	Same working group; internal audit joins.	Same working group.
Continuous education How will learning continue as the technology and the system change?	Refresher when the policy or the vendor's model changes.	Yearly update for audit on new fraud patterns, such as generated receipts.	Update whenever the policy changes.
Conclusion What adoption requires for this project.	Adoption is likely if confirming takes seconds and reviewers are reassured about their jobs.	Transparency towards staff is the precondition.	Easy.

Bring the analysis together and choose a balanced portfolio

RISK-BENEFIT PLOT: write the project letters in the matching quadrant



SWOT PER PROJECT

	A	Receipt auto-fill	B	Fraud & duplicate flags	C	Policy pre-check
Strengths		Data and ground truth available; mature technology; clear Finance ownership.		Builds on A's data; strong support from audit.		Cheap; the rules engine already exists.
Weaknesses		No AI experience; very varied receipt quality.		Few labelled cases; no data scientist.		Small benefit; little learning for later AI projects.
Opportunities		Structured data enables spend analytics (idea 4). A quick win builds trust for B.		Stronger controls; deterrence.		Clarifies a confusing policy.
Threats		Vendor price rises; fake receipts; slow privacy approval.		Bias against some departments; staff distrust; GDPR and works council.		Rules go out of date when the policy changes.
Decision		Prototype now		Prepare first		Implement
Next step, owner, date		8-week prototype with the expense-software add-on and 50 volunteers in all three countries, after privacy sign-off. Owner: Finance director. Go/no-go: January 2027.		Record every flag and its outcome from now on. Consultant analyses 3 years of audit data. Decide up-skill or hire. Owner: head of internal audit. Review: March 2027.		Configure the rules engine with Finance (no AI). Measure the returns rate after 3 months. Owner: Finance policy lead.

ORGANISATION'S RISK STRATEGY

Moderately risk-averse: prefer proven technology, but accept one exploratory project with a limited budget.

PORTFOLIO DECISION: what we do now, next and later, and how funding is released

A balanced portfolio. Now: a low-risk quick win (A) and a non-AI improvement (C). Next: prepare the higher-risk project B, which reuses A's data. Funding is released in rounds: A's prototype results unlock the A contract; B's data analysis decides whether B gets a prototype budget. Idea 4 goes to the BI team.